



Lab-01 Report

Course: CISC5950

Title: Big Data Computing

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Task-1: Suspicious User Detection

We used a two-stage Hadoop Streaming MapReduce workflow to find suspicious users in given dataset to finish Task-01. We started with the wordcount3 sample implementation that was in the GitHub repository. We then changed the template files (mapper-1.py, reducer-1.py, mapper-2.py, reducer-2.py, and test.sh) so that they would look for suspicious users instead of counting words.

In the first step of MapReduce, the mapper was changed so that it could read structured web log entries instead of breaking lines into single words. We looked at each line of the log to find the client's IP address, the time stamp, the HTTP status code, and the request path. The timestamp gave us the hour value (00–23). We counted three things for each request: the total number of requests (req), the number of errors (err, if the status code starts with 4xx or 5xx), and the number of sensitive path accesses (sens). The sensitive paths were /admin, /login, /wp-login.php, /.env, /phpmyadmin, /cgi-bin, and /user_login.php. We sent a composite key in the format hour|ip to make sure that Hadoop Streaming groups things correctly (it only looks at the first tab-separated field as the key). The mapper's output was [hour|ip req err sens]. The reducer in Stage 1 put together all the records that had the same key for hour|ip. It added up the values of req, err, and sens, and then used the formula to find the score of suspicion. The reducer sent out this structured output after figuring out the score [hour ip req err sens score].

In the second stage of MapReduce, we worked on putting suspicious users in order by hour. The second mapper changed the output of Stage-1 by using only the hour as the key and the score and other metrics as the value. This made sure that the reducer put all of the IP addresses that were in the same hour together. The Stage-2 reducer took all the entries for a certain hour, sorted them by suspicion score from highest to lowest, and picked the top three IP addresses with the highest scores. The final output format was [hour ip score req err sens]. This made sure that three of the most suspicious IP addresses were reported every hour.

We changed the test.sh script from the wordcount3 example so that it works with our new logic. The script sets up the Hadoop cluster, puts the log dataset into HDFS (/lab1/input), runs Stage-1 and Stage-2 MapReduce jobs one after the other, and then prints the results from /lab1/output2. The last output was also saved to a local file (task1_output.txt) so that records could be kept and

pictures could be taken. We looked and saw that there were exactly three IP addresses for each hour and that the formula gave the right suspicion scores. For hour 19, one entry looked like this: [19 157.71.254.175 825 74 57 58].

```
root@node-0:/mapreduce-test/mapreduce-test-python/lab1_task1# head -10 task1_output.txt
00      206.24.211.38   705    64     47     50
00      102.189.36.219   673    58     45     48
00      222.169.31.192   669    57     44     48
01      131.103.202.87   733    64     43     54
01      97.168.53.177    711    64     49     50
01      100.247.163.2    653    62     47     45
02      102.194.53.106   720    64     52     50
02      206.81.105.93    658    57     37     49
02      28.40.110.167    617    50     39     45
03      102.189.36.219   744    66     46     54
```

Fig. 1: First 10 records of Task-01 output generated from access-500k.log.txt, showing the top three suspicious IP addresses per hour based on the computed suspicion score.

```
root@node-0:/mapreduce-test/mapreduce-test-python/lab1_task1# head -10 task1_output.txt
00      138.109.145.94   587    54     41     41
00      97.168.53.177     550    54     42     37
00      206.81.105.93    524    48     32     38
01      157.71.254.175   582    51     37     42
01      97.168.53.177    545    47     36     39
01      206.141.158.179  535    50     35     38
02      244.42.26.55     500    44     32     36
02      102.194.53.106   486    43     31     35
02      206.24.211.38    468    47     27     34
03      176.181.14.42    628    54     38     46
```

Fig. 2: First 10 records of Task-01 output generated from access-1m.log.txt, showing the top three suspicious IP addresses per hour based on the computed suspicion score.

Task-2: Time-range “search query”

In Task-02, we added a time range query with parameters to the system that finds users who are acting suspiciously. The goal of this task was to let the program take an hour range argument, like 00-06, and show the three IP addresses with the highest suspicion score for all hours in that range. For Task-01, the best three IPs were chosen for each hour. In Task 02, you had to put the top three IPs in order for the whole time period. We made this change by changing the same two-stage Hadoop Streaming framework that we used in Task 01.

Mapper-1 required most of the modification. We added support for command-line arguments so that the mapper could take a time range parameter in the form of HH-HH. The mapper uses this argument to find the start and end hours. The mapper checks to see if the hour from the timestamp is within the range set for each log entry after getting it. The range follows the lab's instructions, which say to include the start but not the end (for example, 00–06 includes hours 00 through 05). The record won't be counted if the hour isn't in the right range. This filtering makes sure that only the most important log entries are worked on next.

The Stage-1 logic is the same as it was in Task-01 for the rest of the stage. The mapper gets three numbers for each request: the total number of requests (req), the number of errors (err for HTTP 4xx and 5xx status codes), and the number of sensitive accesses (sens for predefined sensitive endpoints). Hadoop Streaming sends out a composite key hour|ip to make sure that things are grouped correctly. The Stage-1 reducer takes all the records for each (hour, ip) pair, adds up the metrics, and then uses the formula to find the suspicion score. After that, it gives you results that are easy to read and show the hour, IP address, combined metrics, and score that was found.

The second MapReduce stage was changed to allow for global ranking over the time range that was chosen. The Stage-2 mapper doesn't group records by hour like Task-01 does. Instead, it gives each record a constant key (ALL). This makes it so that all of the filtered records from the chosen time range have to go to one reducer. The reducer then gathers all of the entries, sorts them in order of suspicion score from highest to lowest, and picks the three IP addresses with the highest scores from the whole range. The final output format is still [hour ip score req err sens]. This method

makes sure that ranking is done on a global scale within the given time frame, not just once an hour.

The test.sh script was changed so that it could take a time range as an argument and send it to mapper-1.py during Stage 1. The script runs both MapReduce stages one after the other and saves the final results in HDFS at /lab1/task2_output2. The results are also sent to a local file called task2_output.txt for record-keeping and taking screenshots. For instance, when we ran "**bash test.sh 00–06,**" the output had exactly three entries from hours 00–05, with the highest score first.

```
root@node-0:/mapreduce-test/mapreduce-test-python/lab1_task2# head -10 task2_output.txt
05      150.65.67.219    885      71      58      64
04      124.65.254.79    788      69      53      56
04      102.194.53.106   785      68      49      57
```

Fig. 3: Global top-3 suspicious IPs identified in Task-02 from access-500k.log.txt after time-range filtering and suspicion score computation.

```
root@node-0:/mapreduce-test/mapreduce-test-python/lab1_task2# head -10 task2_output.txt
04      102.194.53.106   644      54      40      47
05      176.181.14.42    638      55      41      46
03      176.181.14.42    628      54      38      46
```

Fig. 4: Global top-3 suspicious IPs identified in Task-02 from access-1m.log.txt after time-range filtering and suspicion score computation.

Video Demonstration

https://drive.google.com/file/d/1_L-JHM_XXPk0PrSS_yNoIJxWzt41Od/view?usp=sharing